

Teacher:	Ray Joshua Balingkit	**Subject:**	Design & Technology
Week:	1 (Term 1)	**Year:**	10–11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will identify a user need and apply agreed classroom and workshop routines. They will develop their understanding through a teacher explanation, a worked example and an evidence-based classroom task. In the second lesson, students will analyse an everyday product using evidence about function, user needs and material choices. Their recorded reasoning, practical evidence and final checks will show whether they can apply the ideas to a defined user, product or design problem.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher questioning, labelled diagrams, case comparisons and the completed independent tasks. Check accurate terminology, evidence linked to the user or brief and a reasoned explanation of choices. Review recorded tests or traces alongside conclusions, and use the final check in both lessons to identify the next skill needing practice.

LESSON SLOTS - TUESDAY AND WEDNESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	25 August 2026	By the end of this lesson, students will be able to identify a user need and apply agreed classroom and workshop routines.	**H:** Display two pencil holders, one stable and one easily tipped. Ask which better serves a student and what evidence would help decide. Separate a user problem from a preferred colour or shape. **ID:** Introduce designing as responding to a user, purpose and context. Explain the cycle of investigating, proposing, modelling, testing and improving. Establish equipment checks, clear work areas, permission before tool use and reporting damage; this term begins with paper models and teacher-prepared components. **WD:** Model a brief observation of a crowded desk. Record the user, the difficulty reaching stationery and a possible design need without jumping to a finished	**Support:** Provide a user-purpose-problem grid and illustrated routines. Model one observation and allow an oral explanation before writing. Give roles so both partners contribute to the investigation. **Extension:** Students identify a second user with a conflicting need and explain what further research would be needed before choosing a solution.

			solution. Demonstrate safe collection, carrying and return of the materials permitted for today. **YD:** Pairs examine a teacher-prepared desk scenario, record two observations and write a user-need statement. They identify the routines needed for a small card modelling task, organise their workspace and explain how each routine helps the group work safely and efficiently. **C:** Collect the need statements and ask students to distinguish evidence from an assumption about the user. Check that each student can describe what to do when equipment is damaged or an unfamiliar tool is needed.	
2	26 August 2026	By the end of this lesson, students will be able to analyse an everyday product using evidence about function, user needs and material choices.	**H:** Show a reusable bottle and ask whether calling it good design is enough. Invite students to name a feature that can be observed or tested rather than offering a personal preference. **ID:** Explain product analysis through function, user, ergonomics, materials, construction, appearance, cost and environmental considerations. Distinguish an observed feature from a proposed reason for it. Explain that analysis should relate choices to the intended use. **WD:** Annotate a bottle lid, grip and base. Connect the lid seal to preventing leaks and the grip to handling, noting that suitability for a particular hand size requires user evidence. Model a simple non-destructive stability check and record its	**Support:** Offer a labelled product image, an analysis checklist and sentence starters. Limit the first analysis to function, grip and materials before adding the remaining categories. **Extension:** Students explain a trade-off introduced by their improvement, such as greater capacity increasing size, and propose a fair way to test whether the change is worthwhile.

			limitations. **YD:** Students analyse a safe everyday product such as a pencil case or stationery organiser. They sketch and label three features, connect each to a user need and propose one improvement. Partners challenge one unsupported claim and students revise it using observations or a suggested test. **C:** Ask students to justify one feature using the pattern feature, evidence and user benefit. Check that an improvement addresses a stated difficulty and does not simply add decoration.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE (9–1) Design and Technology revision notes: 1.13 Contexts which inform design outcomes; 1.17 Developing and communicating design ideas
- Teacher computer, display screen and whiteboard markers
- Prepared product scenarios, diagrams and user briefs aligned to the lesson objectives
- Prepared analysis worksheet, comparison grid or test-recording table
- Paper, pencils, rulers, card templates and approved classroom stationery
- Teacher-prepared samples or component cards; pre-cut model parts where required
- Lesson success criteria, observation checklist and labelled evidence folders



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?